# Ho Hoang Duy

Hardware Design Engineer
+84 704735647 | hoangduy2206.ute@gmail.com | duyhohoang.github.io | Binh Hung Ward, Ho Chi Minh City

## OBJECTIVE

I'm an Automation Engineering graduate from HCMC University of Technology and Engineering, passionate about electronic hardware design, PCB layout, and embedded systems. Seeking a Hardware Design Engineer position to apply my technical skills, gain practical experience, and contribute to product development.

## TECHNICAL SUMMARY

**Circuit Design and layout in 2 to 4 layer PCB design:**

- Designing AC-DC, DC-DC power supply with various power supply Topology (Buck, Boost, LDO).
- Understand basic signal integrity, EMI/EMC.
- Designing IoT circuits PCB layer with chips like STM32 and ESP32.
- PCB design fundamentals: Stackup configuration, Constraints, Component placement, Signal routing, Routing, and Return paths.
- Designing PCB layer circuits for wireless connectivity LoRa, Wifi, SIM 4G.
- Design high-speed interface: Ethernet, USB.
- PCB design including schematic design, PCB layout, and BOM management.

**Programming Languages and Control System:**

- Embedded firmware development using C/C++ for STM32 and ESP32 microcontrollers.
- Applied communication protocols including I2C, UART, SPI, CAN Bus, RS232, and RS485.

## PERSONAL PROJECT

**IoT Node and IoT Gateway**

- Designed power conversion circuits from 220VAC to 24VDC, 5VDC, and 3.3VDC.
- Integrated SHT31 temperature and humidity sensor via I2C, with RTC-based timekeeping and EEPROM/SD card data storage.
- Implemented RS485 sensor communication and wireless connectivity using Wi-Fi and LoRa.
- Developed relay control circuitry for industrial contactor switching via the IoT Vision APP.

**Industrial Pump Controller & Data Logger**

- Designed an STM32F407 based controller for industrial pump monitoring, VFD control and data logging.
- Developed multi-rail power supplies (12V, 5V, 4V, 3.3V, and 1.8V).
- Integrated peripheral modules including RTC, EEPROM, SD Card, Relays, and Buzzer.
- Designed communication interfaces including 2× CAN Bus, 1× RS485, 2× RS232, Ethernet, and SIM7600/GNSS connectivity.

## TECHNICAL SKILL

**EDA & PCB:** Kicad, Altium, Arduino IDE, STM32CubeMX, PCB Toolkit.

**Programming:** C/C++.

**Testing:** Multimeter, Oscilloscope.

## EDUCATION

**Ho Chi Minh City University of Technology and Engineering** 9/2019 - 9/2026

Automation and Control Engineering Technology

GPA: 2.7/4.0